

Transform Calculus

(MA20202)

Assignment-5

1. Use the Fourier series

$$x^2 = \frac{\pi^2}{3} + 4 \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \cos nx$$

to deduce the value of

$$\sum_{n=1}^{\infty} \frac{1}{n^4}.$$

2. Expand the function $f(x)$, defined by

$$f(x) = \begin{cases} \cos \frac{\pi x}{l}, & 0 \leq x \leq \frac{l}{2} \\ 0, & \frac{l}{2} \leq x \leq l \end{cases}$$

in cosine series.

3. Expand the function $f(x)$, defined by

$$f(x) = \begin{cases} x, & 0 \leq x \leq \frac{l}{2} \\ l - x, & \frac{l}{2} < x \leq l \end{cases}$$

in Fourier series.

4. Find the Fourier series of the function defined as

$$f(x) = \begin{cases} x + \pi, & 0 \leq x \leq \pi \\ -x - \pi, & -\pi \leq x \leq 0 \end{cases}$$

and $f(x + 2\pi) = f(x)$.

5. Find the Fourier half range cosine series of the function defined as

$$f(t) = \begin{cases} 2t, & 0 < t < 1 \\ 2(2 - t), & 1 < t < 2 \end{cases}$$

6. Obtain the complex form of the Fourier series of the function

$$f(x) = \begin{cases} 0, & -\pi \leq x \leq 0 \\ 1, & 0 \leq x \leq \pi \end{cases}$$

7. Prove that for $0 < x < \pi$

$$\text{a. } x(\pi-x) = \frac{\pi^2}{6} - \left[\frac{\cos 2x}{1^2} + \frac{\cos 4x}{2^2} + \frac{\cos 6x}{3^2} + \dots \right]$$

$$\text{b. } x(\pi-x) = \frac{8}{\pi} - \left[\frac{\sin x}{1^2} + \frac{\sin 3x}{3^2} + \frac{\sin 5x}{5^2} + \dots \right]$$

Deduce from a. and b., respectively, that

$$\text{c. } \sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{90}$$

$$\text{d. } \sum_{n=1}^{\infty} \frac{1}{n^6} = \frac{\pi^4}{945}$$

8. Given that $f(x) = x+x^2$ for $-\pi < x < \pi$, find the Fourier series expression of $f(x)$. Deduce that

$$\frac{\pi^2}{6} = 1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \dots$$

9. Find the Fourier series expansion for $f(x) = x + \frac{x^2}{4}$, $-\pi \leq x \leq \pi$.

10. Expand the function $f(x) = x \sin x$, as a Fourier series in the interval $-\pi \leq x \leq \pi$. Hence deduce that

$$\frac{1}{1.3} - \frac{1}{3.5} + \frac{1}{5.7} - \frac{1}{7.9} + \dots = \frac{\pi - 2}{4}.$$

11. Represent the following function by a Fourier sine series:

$$f(t) = \begin{cases} t, & 0 < t \leq \frac{\pi}{2} \\ \frac{\pi}{2}, & \frac{\pi}{2} < t \leq \pi \end{cases}$$

12. Find the Fourier series corresponding to the function $f(x)$ defined in $(-2, 2)$ as follows:

$$f(x) = \begin{cases} 2, & -2 < x \leq 0 \\ x, & 0 < x < 2 \end{cases}$$